ICSMMEP-26 — Konuşma Metni

Eggshell–Agar Biocomposites: Development and characterisation for sustainable interior applications Betül Ünal · Cappadocia University · Mecca, 3–4 November 2026

29 slayt · 2 142 kelime · slayt bütçelerinin toplamı 15:00

Ölçülen süre: 140 kelime/dakika hızında **15:18**, 145'te 14:46. Yani metin hedefin birkaç saniye üstünde — en sondaki "atılacak cümleler" listesinden ikisini çıkarınca tam 15:00'e oturuyor. Provada kendi hızını ölç ve oradan karar ver.

İngilizce paragraflar konuşulacak metnin kendisidir. Köşeli parantez içindeki Türkçe notlar sahne yönergesidir, okunmaz. Sayıları bilerek yazıyla yazdım; rakam görüp duraksamayasın diye.

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1 · Cover — 0:15

Good morning. Betül Ünal, Department of Interior Architecture, Cappadocia University. The panel in the photograph is the subject of this paper: waste eggshell bound in agar. I will give you the protocol, the characterisation, and an argument about how this class of material should be evaluated.

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2 · Background — 0:35

Eggshell is a continuous food-processing by-product, and it is predominantly landfilled. It is roughly a tenth of the mass of every egg. Chemically it is almost pure calcium carbonate, ninety-four to ninety-seven per cent, in a porous protein-bound structure, with a specific gravity below most mineral fillers.

[Alttaki şemaya işaret et]

So it has two possible fates: landfill, or a secondary mineral resource. The case for eggshell as a mineral phase is settled. This paper is about the phase that binds it.

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3 · Prior applications — 0:40

This is the literature the study departs from. Eggshell has been used as a cement replacement in concrete, a stabiliser in laterite brick and in clay, a filler in asphalt, in eco-ceramic tile, in PVC, in polyurethane foam, in ceramic–metal composites, in bone scaffolds, and in wallpaper. The valorisation

is real and well evidenced.

[Elini orta sütuna götür, aşağı indir]

But read the middle column downwards. In every case the mineral is recovered from waste while the binding phase stays cementitious, ceramic or petrochemical. The environmental burden is carried by the binder, and the binder is untouched.

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4 · Agar as binder — 0:40

The binder here is agar, a thermoreversible polysaccharide from red seaweed. It melts on heating and sets on cooling without chemical crosslinking: no reactive hardener, no cure temperature. Its behaviour with glycerol is well mapped. Hernández and colleagues ran thirty-two formulations by design of experiments in twenty twenty-two and released the dataset openly.

But note the scope of that work: unfilled thin films, in uniaxial tension, for packaging. The compositional logic transfers. The form and the loading case do not, and for an interior surface the loading case is bending.

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5 · Research gap — 0:40

Three gaps follow. First, the binding phase is never biological: no study reports an eggshell composite in which both phases are biological and fully biodegradable. Second, agar is characterised only as an unfilled film in tension; as the binder of a mineral-filled cast solid in bending it is not reported. Third, property sets are fragmentary, with compressive, flexural, fire and hygric behaviour in separate studies on separate formulations.

Let me state the contribution precisely, because overclaiming here would be easy. It is not a novel constituent. It is a documented protocol and a single coherent property set.

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6 · Aim — 0:30

The aim, narrow on purpose: document a reproducible protocol for a fully bio-based eggshell–agar composite, and characterise its mechanical, reaction-to-fire and hygric behaviour against published standards.

Four objectives follow. Document the protocol so another laboratory can repeat it. Characterise against standards rather than ad hoc procedures. Position the results against two material families. And derive an evaluation criterion for this material class. The fourth is where I will spend most time.

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7 · Scope — 0:30

Scope, set out explicitly so that nothing I say is read beyond it. Five formulations, three specimens each for the mechanical work, five properties, three standards, ambient cure in one laboratory by one operator, and an envelope of non-loadbearing interior surfaces.

[Alttaki şeride işaret et]

The filled boxes are what was determined. The outlined boxes are what was not: durability, fire classification, life-cycle assessment, structural use. This is characterisation data, not a performance specification.

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8 · Method — 0:30

Four stages, and the order is the method. Produce, recording every parameter. Measure against published standards. Position against two material families. Propose an evaluation criterion derived from the observed behaviour.

This is material-driven in the sense of Karana and colleagues: the properties are measured first and the application envelope is derived from them, rather than the material being chosen to satisfy a form decided in advance.

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9 · Research questions — 0:25

Five questions. The first two are formulation questions. The next two are benchmarking. The fifth asks whether compressive strength is an adequate criterion for this class at all, and it is the one I care most about. Questions one to four are answered by measurement. Question five is answered by argument from those measurements, which is a different kind of claim.

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10 · Hypothesis — 0:30

Two clauses, tested by different parts of the paper. First: the composite can be produced repeatably under ambient conditions, tested by the protocol and the formulation series. Second: its performance as an interior surface is governed by deformation capacity rather than strength, tested by the mechanical results and the benchmarks.

On provenance: the dissertation states a broader hypothesis framed on spatial resilience. This is its materials-engineering component, restated within the scope of the tests reported.

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11 · Material — 0:15

Four constituents: waste eggshell powder, agar, glycerol and water, with a shellac finish. Everything here is one formulation family. The variation in colour and texture is natural pigment and mould geometry, not composition.

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12 · Stage 1 · Mineral phase — 0:35

Stage one. Whole shell from a single bakery, cleaned by hand, heat-treated at one hundred and twenty degrees for twenty minutes, milled, and sieved through a one hundred and fifty micrometre mesh, passing fraction only.

Two points of precision. The particle size was chosen to match the eggshell polymer-composite literature, so that the mineral phase here is comparable with that work. And I call the heat step decontamination and drying, not sterilisation, because sterilisation needs higher temperatures for longer and sterility was not verified.

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13 · Stage 2 · Binder preparation — 0:40

Stage two, drawn as a temperature–time profile. Water and glycerol are pre-mixed before any solids. Agar is added incrementally under continuous stirring; bulk addition forms agglomerates that do not redisperse. The mixture is taken to ninety-five degrees and held until the agar is fully in solution, because if viscosity builds first the gel network is patchy. Then down to eighty while viscosity develops, and the shell goes in near the end; earlier, it sediments. Fifty-five minutes in total.

The set points and the total time are recorded. The ramp rates are not, and stirring was manual.

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14 · Stage 3 · Casting and cure — 0:35

Stage three. Day zero, cast into silicone moulds, because silicone releases without a parting agent and without damaging a part-consolidated specimen. Demoulded on day four, then seven days at twenty-three degrees, shielded from sun and air movement. Day eleven, finished, with three shellac coats. Note there is no thermal input after the fifty-five minute preparation: the entire cure is at room temperature.

[Panele işaret et]

The limitation: the composite shrinks as water evaporates, and post-drying dimensions were not logged, so every dimension I quote is a mould dimension.

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15 · Formulations — 0:35

The formulation matrix. Two things to read in it. In the compressive series, shell-to-binder is exactly two point zero zero in all three mixes. That was not planned as a control, but it means solids concentration is the only variable, and you can see it in the bars: the water band shortens from C1 to C3.

In the flexural series I declare a confound. Glycerol was halved but batch mass was held constant, so water rose to compensate. The plasticiser effect is not cleanly isolated.

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16 · Test programme — 0:30

Five properties, three standards, four specimen geometries, drawn along the bottom. I want the reporting convention on record before any numbers appear. Mechanical values are means of three specimens; individual values were not retained, so I cannot give standard deviations. Ignitability, absorption and density are single determinations, indicative rather than characteristic. Compression was cube compression, not to a named standard, and I flag that as a deviation.

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17 · Compression — 0:35

The compression result, which answers question one. Solids concentration is the only variable, and compressive stress rises linearly with it, from two point five five to four point one zero megapascals, R-squared zero point nine nine.

Three points will fit almost any line, so I call this a hypothesis rather than a law. But the physical reading is clear. Strength is governed by packing density, not by mineral content. The route to a stronger composite is to remove water, not to add shell.

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18 · Compressive failure — 0:35

This is the failure mode. Linear to about one megapascal, then progressive loss of stiffness as the agar network yields and the mineral redistributes. Peak near ten per cent strain, no abrupt load drop, and beyond peak it keeps deforming while holding together.

[Fotoğrafı göster]

The cube has barrelled. It is shorter and wider, and still a single body. Failure proceeds by progressive crushing rather than by fracture. One honesty point: the strain axis assumes ten per cent maximum deformation from observation, not instrumented measurement.

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19 · Flexure — 0:35

Flexure, to ASTM D790. The decisive variable is plasticiser content. At eighteen point eight per cent glycerol the composite deformed without developing measurable resistance. Halving it gives two point three two five megapascals at a modulus of one hundred and ninety-two. The transition is sharp, and it lies between those two values.

Two limitations. The glycerol effect is confounded with solids concentration. And the specimen is kerfed on its tension face while the standard assumes a solid rectangle, so this modulus is a lower bound.

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20 · Flexural deformation — 0:25

I would rather show you this than ask you to take it on the numbers. The beam is deflected well past

where a comparable ceramic would fail, and it has not parted. Above, the kerfed tension face. Below, the same panel arched by hand. Both test programmes gave the same result: substantial deformation with retained integrity.

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21 · Fire, water, density — 0:30

Ignition at around ten seconds under direct flame, self-extinguishing four seconds after removal, with no lateral spread and no dripping. The probable mechanism is endothermic decomposition of the carbonate fraction, diluting the flame front; not verified calorimetrically. This is an ignitability test. There is no Euroclass classification.

Water: zero point seven per cent mass gain over twenty-four hours of immersion, so coating integrity is the limiting factor, not the composite. Density, one point three eight: above the fungal materials, below fired clay.

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22 · Conventional materials — 0:30

First benchmark, against what this would sit alongside in an interior. Let me be direct. One sixth the compressive strength of MDF, one twentieth that of travertine, and travertine is the relevant aesthetic comparison because the material resembles it. It does beat compressed mushroom panel, the only other fully bio-based board here, by a factor of three. But on strength alone it is not competitive with engineered wood or stone, and I will not claim otherwise.

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23 · Bio-based composites — 0:30

Second benchmark, and this is the one that matters. Against the bio-based family the position changes completely. In compression, an order of magnitude above mycelium and hempcrete.

[Logaritmik eksene işaret et]

In stiffness, one hundred and ninety-two megapascals: between soy bioplastic and flax-fibre PLA, inside the LDPE band. Note what that means in process terms. Water-cast, dried at room temperature, no oven, no autoclave, no extruder, and it reaches the stiffness of melt-processed bio-polymers. Wood is still two orders of magnitude above.

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24 · Modulus of resilience — 1:00

This brings me to question five, and the diagram is the argument.

The shaded area under the elastic region is the energy absorbed per unit volume before permanent damage. That area is one half sigma epsilon, and epsilon is sigma over E, so it reduces to sigma squared over two E. Notice where the modulus sits: in the denominator. The low modulus that counted against the material a moment ago is what raises this quantity. For this formulation it is about fourteen kilojoules per cubic metre.

Strength records the load at which a material stops performing. It does not record whether it then fails gracefully or catastrophically, and for a demountable non-loadbearing component that is what governs service life.

The limitation: I cannot give you a comparative table, because the literature reports strength and modulus separately. This is a data point and a proposed reporting convention, not a ranking.

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25 · Application envelope — 0:30

The envelope follows from the measurements. Non-loadbearing, because four point one megapascals supports nothing more. Dry interiors, because water performance depends on the coating. Away from ignition sources: self-extinguishing, but unclassified. Modular rather than sheet, because damage is then replaced one unit at a time.

Every constituent including the shellac is biodegradable, and reversible fixing recovers the module intact. No life-cycle assessment was carried out, so these are qualitative potentials, not quantified claims.

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26 · Limitations — 0:25

The limitations, in three classes, because the grouping is informative. Measurement: no dispersion data, strain assumed, shrinkage unquantified. Process: the flexural series confounded, the kerfed section calculated as solid, manual stirring in one laboratory by one operator. Untested: durability, fire classification, life-cycle assessment.

I would rather set these out than have them found. Each one corresponds to a defined next experiment.

27 · Further work — 0:25

Four experiments, in order of what each one resolves. Repeat with individual values retained, so dispersion can be reported. Record post-drying geometry, which turns every lower bound here into a true value. Instrument the strain, which makes the modulus of resilience defensible. And test an un-kerfed section, which separates geometry from material.

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28 · Conclusions — 0:30

Five conclusions. A protocol that can be repeated, parameter by parameter. Packing density, not mineral content, governs compressive strength. Plasticiser content sets a brittle or a ductile response, and the transition is sharp. A water-cast, ambient-dried material reaches bio-polymer stiffness without thermal processing. And strength alone is an insufficient criterion for this class: deformation capacity governs service behaviour, so strength and modulus should be reported together.

Thank you. I am happy to take questions.

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29 · References — 0:05

[Slaytı açık bırak, bir şey söyleme. Soru gelirse kaynağı buradan göster.]

Soru–cevap

Kısa tut. Sınırlılığı kabul et, sonraki deneyi söyle, sus.

"Standart sapma yok, nasıl güveneceğiz?" You are right, and I said so before the numbers. Three specimens per formulation, individual values not retained. Repeating the series with dispersion reported is the first item on my further-work slide.

"Basma testi hangi standarda göre?" Cube compression, not to a named standard, and I flagged that as a deviation. Flexure follows ASTM D790-17, ignitability ISO 11925-2, absorption ASTM D570.

"Ölçüler kalıp ölçüsü, gerçek gerilme değil." Correct. The composite shrinks as water evaporates and post-drying dimensions were not logged, so every value I reported is a lower bound. Recording the

dried geometry is the second item on the further-work slide.

"Yivli numuneden modül hesaplanmaz." Agreed, which is why I called it a lower bound rather than a material constant. Testing an un-kerfed section is the fourth item on that slide.

"R^2 = 0.99 ama üç nokta var." That is exactly why I called it a hypothesis rather than a law. What I claim is the physical reading: packing density, not mineral content, governs strength.

"Dirençlilik modülü için karşılaştırma yok." No, and I cannot assemble one, because the literature reports strength and modulus in separate studies on separate formulations. I am proposing a reporting convention, not a ranking.

"Nem, küf, yaşlanma?" Untested. Durability of any kind is outside the scope of this paper and is named on the limitations slide.

"Ölçeklenebilir mi?" I have no data on that. What I can say is that the protocol needs a hotplate, a mould and ambient drying. No oven, no autoclave, no extruder.

"Neden agar? Nişasta ya da jelatin olmaz mıydı?" Thermoreversible setting without a hardener, and it tolerated sixty-six per cent mineral loading without a coupling agent. Comparing binders is a separate study, and a fair one to ask for.

Provaya dair

- Yüksek sesle, kronometreyle oku. Metin olduğu gibi 15:18; hedef 15:00.

Aşağıdaki listeden ilk ikisini çıkarınca tam oturuyor. Uzun kalırsan **hızlanarak değil, cümle atarak kısalt.**

- Atılacak cümleler, bu sırayla:
- 1. Slayt 16, son cümle — *"Compression was cube compression… deviation."*

(Sorulursa zaten cevaplıyorsun, soru–cevap sayfasında hazır.)

- 2. Slayt 22, *"It does beat compressed mushroom panel…"* cümlesi.
- 3. Slayt 7, son cümle — *"This is characterisation data, not a performance

specification."*

- 4. Slayt 25, son paragraf.
- 5. Slayt 27'de dört deneyden ikisini say, dördünü sayma.
- Dokunulmayacak slaytlar: **3, 5, 13, 15, 17, 19, 24.** İskelet bunlar.

- Slayt 24 sunumun tepe noktası. Oraya kadar tempoyu koru, orada yavaşla.
- Köşeli parantezli yönergelerde gerçekten ekrana dön ve işaret et; o üç saniyelik

duraklar metnin ritmini kuruyor.